

ERECT: Evidence Refinement Enhanced Complex Claim Verification with Large Language Models

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Abstract—Automatic claim verification aims to predict the veracity of claims automatically. Large Language Models (LLMs) have been leveraged to check the veracity of claims and achieved great progress, due to their powerful logical reasoning ability and rich knowledge reserve. However, since the LLMs have factual knowledge boundary problem that comes from outdated and incomplete pre-training data, they may experience hallucinations when verifying emerging claims. Evidence retrieval is helpful in improving the performance of LLM-based fact verification. However, the quality of retrieved evidence directly affects the verification effectiveness of LLMs. In this paper, we present ERECT, an Evidence Refinement Enhanced complex Claim verification model, which refines more accurate evidence with the help of LLMs to further assist LLMs in verifying complex claims more accurately. Specifically, ERECT first decomposes complex claim verification into several simpler sub-tasks by LLMs, as the complex claim itself has intricate multi-hop relationships among inside entities, which is not conducive to evidence retrieval and veracity reasoning. In order to conform to practical application scenarios, we assume that the evidence can only be retrieved from some open domain sources, rather than annotated manually. Some common information retrieval method is first adopted to get the coarse-grained candidate evidence set from the open domain sources for each sub-task. Then LLMs are adopted to analyze complex relationships between candidate evidence and sub-task, and select the most useful evidence for each sub-task from candidate evidence. The refined evidence is further used to assist LLMs in executing each sub-task. The final prediction and explanation of the original complex claim are obtained after executing all sub-tasks. Experiments on the real-world datasets HoVER, demonstrate the superiority of ERECT over several baselines on complex claims verification.

Index Terms—fact verification, information retrieval, large language model.

I. INTRODUCTION

The vigorous development of the Internet community has broadened our horizons, yet it also presents challenges with the proliferation of fake news and misinformation. Recognizing the substantial harm misinformation can inflict on society, automatic fact verification (FV) has become a research hotspot in the field of natural language processing (NLP) [1].

The existing FV methods mainly use various pre-trained language models (PLMs) to capture the language patterns of

the claim itself or to mine the differences between the claim and the related evidence to determine its veracity [1]. The development of Large Language Models (LLMs) gives the traditional NLP task a brand new perspective and do the same to the FV problem [2]–[4]. Equipped with the powerful semantic understanding ability and logic reasoning ability, LLMs have achieved promising progress in FV, enhancing both the checking ability and the interpretations [3]. However, for some emerging claims, LLMs may experience hallucinations, due to the limited fact knowledge boundary brought by its outdated and incomplete pre-training data. Therefore, in order to improve the performance of LLM for fact-checking, it is also crucial to have an effective evidence retrieval enhancement.

The quality of the retrieved evidence directly affects the performance of various FV models, including LLM-based models [5], [6]. In order to focus on the research of the FV model itself, early researchers built FV datasets with gold evidence, such as HoVER [5], FEVEROUS [7], TabFact [8], etc. The gold evidence in these datasets is usually retrieved and annotated in advance by human experts. In real FV scenarios, however, emerging claims don't always have gold evidence [1], [9]. During the fact-checking process, appropriate evidence needs to be extracted simultaneously from external sources. In view of the importance of evidence, many works have proposed a variety of evidence retrieval methods for FV. Martin Fajcik et al. [10] present Claim-Dissector, an interpretable fact-checking system with joint re-ranking and veracity prediction to optimize the evidence retrieval process. FOLK [11] introduces open domain search via search engine, like Google, to retrieve external evidence for FV. Most of them mainly rely on search engines or traditional information retrieval methods based on relation computation on vector space model to sift evidence from a large amount of information. However, the extensive use of search engines' APIs incurs substantial costs, and discerning the most reliable evidence from their returned results remains a formidable challenge. Moreover, the traditional information retrieval methods mainly consider the semantic similarity between information and only return information semantically similar to the claims. They can not directly measure the effect of information on veracity prediction to determine the most useful evidence for claims.

To address the above issues, in this paper, we propose

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an evidence refinement enhanced complex claim verification model called ERECT, which refines evidence from a large external corpus to improve the verification of complex claim with the help of a LLM. Specifically, ERECT first decomposes the complex claim verification into simpler program steps with straightforward logical relationship by a program generation method, as the intricate multi-hop logical relationships in the complex claim is not conducive to evidence retrieval, and one piece of evidence can usually only support the verification of one straightforward logical relationship [12], [13]. The large evidence corpus, such as Wikipedia, typically contains billions of sentences. Traditional information retrieval method, such as BM25 [14], [15], is adopted to search the text blocks that are related to each program step to form the coarse-grained candidate evidence set for each program step. As the information retrieval method cannot judge which sentence can directly assist the execution of each program step, the candidate evidence set should contains all related sentences potential to be evidence, which inevitably includes many redundant sentences useless for step execution. Thus we further refine the evidence from the candidate evidence set for each step with the help of LLM, as it has powerful semantic understanding and logic reasoning ability to judge the most appropriate evidence for each step. The final veracity label and verification explanation for the original complex claim is got after the execution of all program steps. The generated explanations can aid human experts in checking the model's verification process. ERECT is mainly tested on several subsets of a challenging dataset HoVER with complex claims [5]. The results demonstrate that in the verification of highly complex claims, ERECT surpassed several other competitive baselines.

The contributions of this paper are as follows:

- Introduction of ERECT: We propose a novel evidence refinement enhanced complex claim verification model, ERECT, which effectively decomposes complex claim verification into simpler steps and get refined evidence to support the execution of simpler program steps.
- Integration of LLM for evidence refinement: Our approach leverages large language models (LLMs) to refine evidence from a large external corpus, ensuring that the most relevant evidence is selected in evidence retrieval step. This refinement significantly enhances the precision of the claim verification process.
- Evaluation of the importance of evidence: We designed ablation experiments to test the performance of the same model under different evidence type settings, quantifying the importance of evidence accuracy in the FV task.

II. RELATED WORK

1) *Fact Verification*: Fact Verification aims to automatically predict the veracity of claims. Some works [16]–[18] did not use any external evidence and focused on the claims themselves, which verify claims by exploiting their language patterns, including textual content, stylistic nuances, and a

variety of syntactic and semantic elements. However, the language pattern of claim is easily deliberately forged to deceive the verification model [19]. Especially with the development and application of LLMs, the problem has been further exacerbated, as LLMs may be maliciously used to generate claims that appear trustworthy but contain misleading information [2], [20]. Therefore, in order to enhance the reliability of FV, more and more models [10], [21] rely on external evidence to determine the veracity of claims. They model the relationship between claims and evidence through various semantic models and attention mechanisms to determine whether claims are supported or refuted by evidence [19].

The LLMs with powerful semantic understanding and logic reasoning ability, as well as rich knowledge reserve have achieved promising progress in many NLP tasks, including FV task. Liangming Pan et al. [22] proposed ProgramFC which decomposes complex claims into simpler sub-tasks by generating reasoning programs with the help of in-context learning ability of LLMs. Each program step can be executed by a shared library of specialized functions, and the final prediction result is got after all steps are executed. Given the effectiveness of complex claim decomposition in improving verification performance, FOLK [11] also adopts a First-Order-Logic (FOL) clause to assist in the claim decomposition and introduces open domain search like Google to retrieve external evidence. These exquisitely designed LLMs-based FV methods have achieved significant performance. After carefully analysis, it is found that the performance bottleneck of these methods is mainly limited by the quality of evidence retrieved. Thus, we propose a evidence refinement method to eliminate the performance bottlenecks of these LLMs-based FV methods.

2) *Evidence Retrieval*: Relevant work has demonstrated that retrieving additional evidence can improve the performance of fact verification [23]. The typical evidence retrieval techniques are based on traditional information retrieval methods which retrieve the top-k relevant passages related to claims from a large corpus by setting the claims as query conditions [19]. Specifically, for evidence retrieval enhanced fact-checking, most of the existing methods [22], [24] employ a 3-stage pipeline. The multiple steps of the verification process may lead to error propagation and the top-k retrieved evidence may not be able to either support or refute the claim. Therefore, some previous works [25] focused on merging evidence retrieval and veracity prediction into a single model to reduce the impact of error propagation. For example, Claim-Dissector [10] introduces a joint approach which jointly identifies the relevant evidence to the given claim and predicts the veracity of the claim. Despite its improved performance, it may still suffer from the adverse impact of irrelevant information and fails to eliminate redundant and irrelevant evidence retrieved by BM25 algorithm. To tackle these problems, we propose to further refine the evidence by employing LLM to select the most useful evidence, thereby eliminating irrelevant information and ensuring the evidence is exactly what the claim needs.

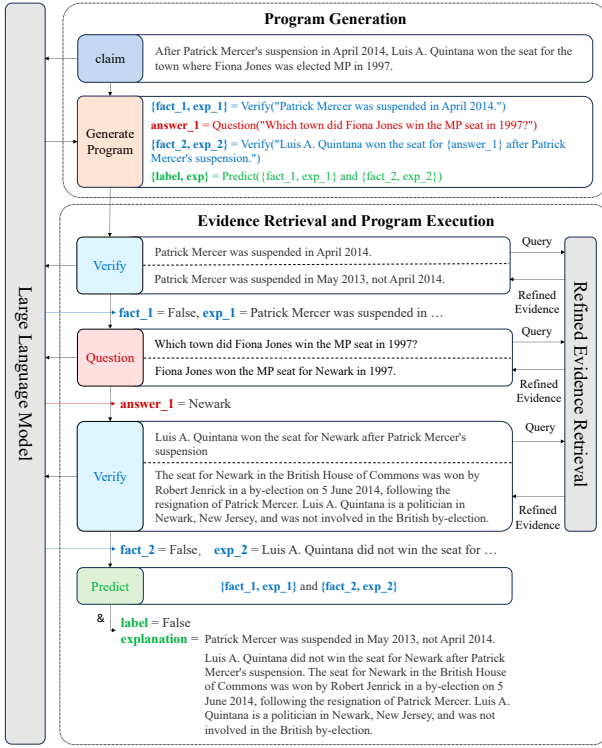


Fig. 1. Overview of model ERECT

III. METHOD

The claim verification task can be formulated as given a claim C , On the basis of a knowledge source E , the model F predicts its veracity label Y and gives an explanation Exp .

As illustrated in Fig. 1, the overall framework of ERECT consists of two modules, i.e., *program generation*, and *evidence retrieval and program execution*. In the *program generation* module, inspired by ProgramFC [22], a complex claim is first decomposed into simpler program steps with straightforward logical relationship based on LLM to simplify the difficulty of evidence retrieval and veracity prediction. The *evidence retrieval and program execution* module employs a Refined Evidence Retrieval model. It first generates a candidate evidence set using traditional methods like BM25 on large corpora like Wikipedia. Then, it refines the set by leveraging LLM to select the most relevant evidence, eliminating redundant sentences that don't aid in program step execution. All program steps are executed based on their refined evidence to get the final veracity label and verification explanation for the original complex claim. The generated explanation can aid human experts in checking the model's verification process, enhancing its transparency and interpretability. Each of module is described in details next.

A. Program Generation

A complex claim C contains multiple entities and intricate multi-hop logical relationships that occur between them. The complex logical relationships bring difficulties to both

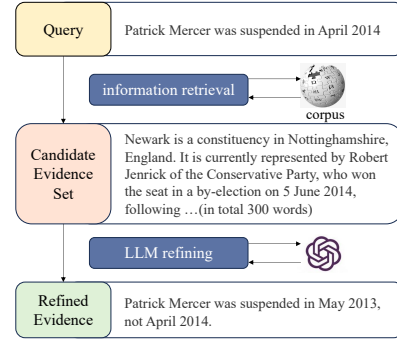


Fig. 2. The refined evidence retrieval model

evidence retrieval and fact verification, as it is hard for the models to understand the complicated semantics in complex claims and retrieve appropriate multiple pieces of evidence to jointly verify the claims. Usually, one piece of evidence can only support the verification of one straightforward logical relationship. Therefore, in order to simplify the difficulty of evidence retrieval and veracity prediction, inspired by ProgramFC [22], ERECT first decomposes the verification of original complex claim into a set of simpler program steps $P = [S_1, \dots, S_n]$ with straightforward logical relationship, which is called program generation. Each step $S_i \in P$ is an instruction in controlled natural language that guides the corresponding sub-task function, which may be implemented with the help of LLM. Formally, each program step can be defined as $S_i = (f_i, A_i, V_i)$, where f_i is the sub-task function defining a operation need to be performed in the task, A_i is the argument passed to the function, and V_i is the variable that stores the returned result from the function execution $V_i = f_i(A_i)$. As it is common for a subsequent step to depend on the results from previous steps, the argument of a step A_i can refer to the results of the preceding step, i.e., $V_1, \dots, V_{i-1}$. The allowed sub-task functions include *Question* which takes a question Q as the input argument and returns the answer A to the question, *Verify* which takes a claim C as the input argument and returns a veracity label of either TRUE or FALSE as well as its explanation, and *Predict* which is usually the last program step that takes a logical expression over the previously returned variables as input and output a boolean value indicating the veracity label of the original claim C , as well as its verification explanation. Similar to ProgramFC, the program is generated by prompting the LLM with an instruction of the task, K in-context examples, and the input claim C .

B. Evidence Retrieval and Program Execution

In order to support the execution of *Question* and *Verify* function as precisely and briefly as possible, a refined evidence retrieval model is designed to select the most appropriate evidence from a large evidence corpus for each program step except the last *Predict* step. As illustrated in Fig.2,

since the large evidence corpus, such as Wikipedia, typically contains billions of sentences and is hard to locate the most appropriate evidence directly, traditional information retrieval method, such as BM25, is adopted to search the text blocks related to each program step to form its candidate evidence set by setting the the argument of each program step as query. However, the information retrieval method based on relation computation on vector space model usually cannot judge which related sentence can help to execute each program step directly. So the candidate evidence set should contains all related sentences potential to be evidence, which inevitably includes many redundant sentences useless for step execution. On the other hand, the powerful semantic understanding and logic reasoning ability of the LLM makes it possible to judge the most appropriate evidence for each program step directly. Thus we further prompt the LLM to refine the evidence from the candidate evidence set to assist the execution of each step. The prompt used for evidence refinement is as follows:

Given the claim and the evidence set, select the most relevant evidence that can be used to support or refute the following claim. Evidence that supports a claim is information that is consistent with the content of the claim. And evidence that refutes a claim is information that contradicts the facts in the claim. Return the refined evidence as directly as possible.

Claim: {CLAIM}

Evidence set: {EVIDENCE}

The evidence after selection is:

After the evidence refined, each program step is also executed by prompting the LLM. Specifically, the basic pattern of *Question* is to input the refined evidence E_i and target questions A_i into LLM, and then get a phrasal answer V_i that is used to supplement other steps, i.e., $answer_i = Question(A_i, E_i)$. The prompt for *Question* is as follows:

Given the question and the evidence where the answer can be found, return the answer directly with no explanation.

Evidence: {EVIDENCE},

Question: {QUESTION},

The answer is:

The *Verify* operation synthesizes the decomposed claim A_i with the retrieved evidence E_i to predict veracity label as well as generate verification explanation, i.e., $[fact_i, exp_i] = Verify(A_i, E_i)$. The prompt for *Verify* is as follows:

Given the claim and the evidence, judge whether the claim is supported or refuted by the evidence, return the answer 'True' or 'False' with explanation.

Evidence: {EVIDENCE},

Is it true that {CLAIM}? True or False?

The answer is:

The final *Predict* operation is the execution of a logical expression over variables returned by previous *Verify* operations, and obtains the boolean value indicating the veracity label of the original claim, as well as its verification explanation by combining all previous returned explanations.

IV. EXPERIMENTS

A. Datasets

ERECT focuses on complex claim verification that requires multi-step reasoning combined with multiple pieces of evidence. To test this performance, We select HoVER [5] to evaluate ERECT. HoVER is created specifically to test a model's ability to perform multi-hop fact checking on complex claims, and contains a variety of claim types. The fact verification of each claim requires reference to Wikipedia articles of two to four hops. Based on the number of hops, the datasets are divided into 3 subsets: 1,126 two-hop claims, 1,835 three-hop claims, and 1,039 four-hop claims.

B. Baselines

We compare ERECT to six baselines categorized into three groups.

- 1) Pre-trained models. The pre-trained models are leveraged for FV directly. BERT-FC [26] leverage BERT for FV. LisT5 [24] leverage T5 for FV.
- 2) Fine-tuned models. They are pre-trained models fine-tuned on other FV or natural language inference (NLI) datasets. RoBERTa-NLI [27] fine-tune RoBERTa-large on four NLI datasets. MULTIVERS [28] is a LongFormer [29] model fine-tuned on FEVER.
- 3) LLM-based models. They are the models adopting LLM for FV. FOLK [11] incorporates a First-Order-Logic (FOL) clause to facilitate claim decomposition and integrates open domain search Google for retrieving external evidence. ProgramFC [22] decomposes complex claims into simpler sub-tasks by generating reasoning programs and predicts veracity by program execution. We select N=1 for ProgramFC for fair comparison, i.e. it doesn't use aggregating reasoning.

C. Experimental Settings

For evidence retrieval setting, We use BM25 [14] implemented with the Pyserini toolkit [15] as the information retriever for both ERECT and the baselines, and adopt the Wikipedia corpus as the knowledge source. We uniformly retrieve the top-5 paragraphs as the candidate evidence set for ERECT.

On the choice of LLM, gpt-3.5-turbo is used uniformly as a working LLM for all LLM-based methods to erase the differences between LLM performance. Previously, some methods, such as FOLK and ProgramFC, used the text-davinci-003 (175B) model known for prompt engineering. However, as

TABLE I
MACRO-F1 SCORE OF ERECT AND BASELINES ON DATASET HOVER.
THE BEST RESULTS ARE IN BOLDFACE AND THE SECOND-BEST RESULTS
ARE UNDERLINED.

Model	HoVER (2-hop)	HoVER (3-hop)	HoVER (4-hop)
BERT-FC [26]	50.68	49.86	48.57
LisT5 [24]	52.56	51.89	50.46
RoBERTa-NLI [27]	63.62	53.99	52.40
MULTIVERS [28]	60.17	52.55	51.86
FOLK [11]	68.67	53.25	48.35
ProgramFC [22]	64.23	<u>56.48</u>	<u>52.87</u>
ERECT	<u>65.91</u>	58.77	53.76

gpt-3.5-turbo is a more commonly used and cheaper model now, we choose it as the working LLM for all methods for fair.

In order to improve the performance, a few in-domain examples can be used for fine-tuning the pre-trained models (BERT-FC and LisT5), for continuous fine-tuning the FC/NLI fine-tuned models, or as in-context examples for the LLM-based models. Therefore, for a fair comparison, we restrict all models to have access to only 20 in-domain examples from HoVER for few-shot learning. By providing the same number of examples, we can ensure that the amount of information obtained by all models in the few-shot learning phase is consistent.

D. Main Results

Macro-F1 score is used as the evaluation metrics in all experiments. Table 1 shows the performance of ERECT and other baselines on dataset of 2-hops, 3-hops and 4-hops. The experimental results show that ERECT is superior to most FV baselines, especially when the claims have more hops and are more complex. Specifically, the following observations can be made from the experimental results.

(1) It can be observed that the more hops the claims have, i.e. the more complex the claims are, the worse the performance of various models is, which is consistent with the theory that the more complex claims are more difficult to be verified.

(2) Fine-tuned models perform better than pre-trained models, which indicates that fine-tuning on some task-related datasets indeed improves the performance. LLM-based models outperform fine-tuned and pre-trained models in most cases, demonstrating the strong ability of LLM in analyzing, decomposing and verifying complex claims [11], [22].

(3) The proposed ERECT has the best performance on 3-hop and 4-hop datasets which are more complex than 2-hop dataset, and is only slightly worse than FOLK on 2-hop dataset. This maybe because that when the claim is simpler, it is easier for traditional information retrieval methods to retrieve related information that can help to judge the claim, while the refined evidence retrieval model in ERECT may eliminate some important information by further refinement operation. As the claim becomes more complex with more hops, the traditional information retrieval methods may be

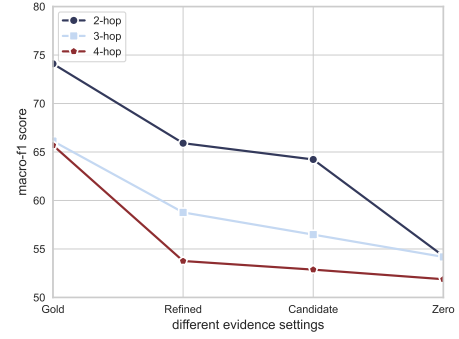


Fig. 3. The Impacts of Accurate Evidence

powerless in retrieving accurate evidence, while the refinement operation in ERECT will take effect.

E. Impacts of Refined Evidence

To gain more insights on the importance of refined evidence, we perform an ablation study on the HoVER dataset. Our goal is to evaluate how changes in evidence affect the performance of the model. Specifically, We defined four types of evidence setting for ERECT:

- **Zero Evidence.** In this setting, external knowledge source is empty $E = \emptyset$, i.e. no external evidence is included for FV.
- **Candidate Evidence.** All candidate evidence in the candidate evidence set are used in each program step without evidence refinement operation.
- **Refined Evidence.** This evidence setting is just the ERECT's setting.
- **Gold Evidence.** The gold evidence provided by the HoVER dataset itself is adopted in each program step. It is worth noting that gold evidence is usually manually annotated.

The experimental results of different evidence settings are reported in Fig. 3. It is shown that the accuracy of evidence affects the results of claim verification. When there is no evidence introduced, the model performs worst. It suggests that the evidence is indispensable in FV. The quality of evidence also has a significant impact on verification performance. The quality of gold evidence annotated by experts is the best, therefore the performance of the model using it is also the best. The candidate evidence retrieved by traditional information retrieval methods without refinement may contain a large amount of misleading and irrelevant information, which may interferes with the model's judgment, making the model's performance worse than that adopts refined evidence, which is the most appropriate evidence selected from the candidate evidence set by a LLM. This demonstrates the importance of evidence refinement operations and also indicates that evidence optimization is a promising direction for improving the performance of FV models

V. CONCLUSION

In this paper, we propose ERECT, which refines more appropriate evidence from external source with the help of LLMs to further assist LLMs in making more accurate judgement on veracity of complex claims. The model is equipped with good performance because: (i) it first decomposes a complex claim into several simpler sub-tasks through a program generation process, so that the simple logic in sub-tasks facilitates the retrieval of appropriate evidence; (ii) it retrieves the coarse-grained candidate evidence set from open domain source first, and adopts LLM to analyze complex relationships between candidate evidence and sub-task to select the most appropriate evidence for each sub-task. The extensive experiments and ablation studies on real-world datasets demonstrate the superiority of ERECT over several competitive baselines.

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